GRPO

$$r_{\text{sum}}^{(i,j)} = r_1^{(i,j)} + \cdots + r_n^{(i,j)}$$

$$A_{\text{sum}}^{(i,j)} = \frac{r_{\text{sum}}^{(i,j)} - \text{mean}\{r_{\text{sum}}^{(i,1)}, \ldots, r_{\text{sum}}^{(i,G)}\}}{\text{std}\{r_{\text{sum}}^{(i,1)}, \ldots, r_{\text{sum}}^{(i,G)}\}}$$

Notation

r: reward
A: advantage
n: number of objectives
G: number of rollouts
B: batch size
i: i^{th} question
j: j^{th} rollout
$D_{\text{Batch}} = \{\text{question } i \mid i = 1, \ldots, B\}$

Decoupled Normalisation

GDPO

Batch normalisation
to keep a stable numerical range

$$A_1^{(i,j)} = \frac{r_1^{(i,j)} - \text{mean}\{r_1^{(i,1)}, \ldots, r_1^{(i,G)}\}}{\text{std}\{r_1^{(i,1)}, \ldots, r_1^{(i,G)}\}}, \quad \cdots \quad A_n^{(i,j)} = \frac{r_n^{(i,j)} - \text{mean}\{r_n^{(i,1)}, \ldots, r_n^{(i,G)}\}}{\text{std}\{r_n^{(i,1)}, \ldots, r_n^{(i,G)}\}},$$

$$A_{\text{sum}}^{(i,j)} = A_1^{(i,j)} + \cdots + A_n^{(i,j)}.$$

$$\hat{A}_{\text{sum}}^{(i,j)} = \frac{A_{\text{sum}}^{(i,j)} - \text{mean}\left\{A_{\text{sum}}^{(i',j')} \mid i' \in D_{\text{Batch}}, j' = 1, \ldots, G\right\}}{\text{std}\left\{A_{\text{sum}}^{(i',j')} \mid i' \in D_{\text{Batch}}, j' = 1, \ldots, G\right\} + \epsilon}$$